

Muhammad Abdullah | Software Engineer

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EDUCATION

FAST NUCES (Bachelors in Computer Science)
Govt. Graduate College of Science (F.Sc. Pre Engineering)

Sep 2022 – Dec 2026
Aug 2020 – Aug 2022

TECHNICAL SKILLS

Programming Languages: C++, Python, MySQL, MongoDB, JavaScript, PHP

Web & Technologies: HTML5, CSS3, React.js, Node.js, Express.js, Django.js

Platforms: WordPress, Shopify, Git, GitHub, Docker, Container and Kubernetes

PROJECTS

BioMechAI | Final Year Project | Flutter, MediaPipe, Random Forest, Flask, React, Firebase

- Built a real-time AI fitness trainer tracking 33 body landmarks at ~30 FPS using MediaPipe, with a custom scikit-learn Random Forest classifier recognizing 7+ exercises across 1,050+ training videos at ~85% confidence.
- Developed the Flutter mobile app with automatic exercise detection after first rep, real-time rep counting, weight logging, and Firebase-synced session recording with instant trainer feedback notifications.
- Engineered a React-based trainer dashboard enabling certified trainers to review AI-annotated sessions, analyze per-rep correctness, filter by date/muscle group, and provide direct feedback with instant mobile notifications.
- Deployed a Flask backend on Render extracting 21 joint-movement features from pose landmarks and serving classification via REST API; architected full Firebase infrastructure (Auth, Firestore, Storage, Cloud Messaging) for end-to-end cloud sync across 3-member team.

Tensor Film | Smart Movie Explorer | MERN Stack | Next.js | [Open in Browser](#)

- Developed a full-stack AI-powered movie recommendation platform using MERN Stack and Next.js with a responsive and modern user interface.
- Integrated AI to recommend movies based on users' moods and preferences, enabling personalized movie discovery with trailers and detailed movie information.
- Built the frontend using Next.js, TypeScript, and the App Router, ensuring fast performance and a smooth user experience.
- Deployed the application on Vercel with automatic CI/CD through GitHub, while hosting the backend on Render.

Neural Storyteller | Image Captioning Pipeline | ResNet50 + LSTM

- Built an image-to-text pipeline using a Show-and-Tell-style encoder-decoder: pre-trained ResNet50 extracts a 2048-dim feature vector, decoded by a multi-layer LSTM into word-by-word captions.
- Trained on Flickr30k (31,000 images, 5 captions each) using teacher forcing; implemented Beam Search (k=5) at inference to avoid generic/repetitive output.
- Deployed via Gradio for real-time caption generation from user-uploaded images; model distinguishes fine-grained actions and identifies complex objects after 30 epochs.

ACHIEVEMENTS

IBM Full Stack Software Developer | Professional Certificate, Coursera | [View Credential](#)

- Front-End & Back-End Development: React, Node.js, Express.js, Django, Flask, SQL & database management.
- Cloud & DevOps: Docker, Kubernetes, OpenShift, Microservices & Serverless computing, CI/CD.
- Foundations & AI: Git & GitHub, Python for Data Science, Generative AI application development, and a full-stack capstone project.

INTERESTS

Chess, Fitness Training, Gaming, Continuous Learning in Software Development